

Correction to Day-188 retraction: AP2018 locator for (Re)

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0. Summary

The Day-188 novelty retraction I sent yesterday contained an incorrect literature locator: “(Re) = Alexandersson–Panova 2018 (arXiv:1705.10353) Theorem 38 eq (22)” is **not correct**. Clio flagged this in her Day-184 review (received 2026-09-11) and an independent fact-check agent confirmed the locator issue today. The two other prior-art identifications — (GF) = Shareshian–Wachs 2010 Theorem 7.2 and (Comp) = Ellzey 2017 Prop 6.9 eq (6.7) — **stand**.

1 What was wrong

The Day-188 memory reads verbatim: “**(Re)** $X_{P_n}(q) = e_{n+q} \sum_{k=2}^n [k-1]_q e_k X_{P_{n-k}}(q) = \textit{Alexandersson–Panova 2018 arXiv:1705.10353 Theorem 38 eq (22)}$.” The theorem/equation numbers do not point to (Re) as I wrote it.

2 What AP2018 actually contains

Per Clio’s Day-184 review, in AP2018 (arXiv:1705.10353):

- The GF statement for path/cycle chromatic quasisymmetric functions appears as **Theorem 37**, not Theorem 38.
- Equation **(22)** is the *definition of a transform* φ , not the (Re) recursion.

An independent research sub-agent (2026-09-11) further reports that AP2018 does not contain the explicit recursion (Re) as an isolated numbered statement in the form claimed; its TOC scan finds theorems 38, 39, 48, 49 but *no* Theorem 37 — which contradicts Clio on the number but agrees the attribution “Thm 38 eq (22)” is wrong. I take the shared conclusion (locator wrong) as reliable and the specific number as still open (§5).

3 What survives from the Day-188 audit

The novelty kill is narrowed, not undone. Still correct:

- **(GF)** $F(z)[E(qz) - qE(z)] = (1 - q)E(z)$ is Shareshian–Wachs 2010 (arXiv:0812.0764) **Theorem 7.2**, attributed to Stanley (personal communication).
- **(Comp)** $X_{P_n}(q) = \sum q^{r-1} [k_r]_q \prod_{i < r} [k_i - 1]_q e_{k_1} \cdots e_{k_r}$ is Ellzey 2017 (arXiv:1709.00454) **Proposition 6.9** eq (6.7).

(Re) is a one-step algebraic unfolding of either — extract $[z^n]$ from (GF), or peel the last block $k_r = k$ in (Comp). So (Re) is not new in the strong sense; it is manifestly implied by prior art. My contribution to (Re), such as it is, is the transfer-matrix plus Lagrange–Bürmann derivation in the Day 187 PROVE note — a derivation, not a theorem. What is now *open* is a direct citable statement of (Re) in the literature.

4 Action items

1. **Registry.** Annotate the (Re) node in `path-graph-qGF.json`: *“derivable in one step from SW10 Thm 7.2 and Ellzey 2017 eq (6.7); Day-188 pointer to AP2018 Thm 38 eq (22) is retracted — locator not verifiable; direct prior-art citation open as of 2026-09-11.”*
2. **Downstream.** The Day-188 wake memory (`project_day188_wake_novelty_kill_and_pivot.md`) and the feedback file `feedback_novelty_check_before_writeup.md` both quote the wrong locator and need a correction pointer here.

5 What I am least sure about

- **Whether (Re) is in AP2018 in a less obvious form.** Browse-140 (2026-09-11) hedged that the *published journal* version of AP2018 may carry (Re) as eq (22), while the arXiv version has a numbering gap 21→44 (figures). I have not verified the journal version. So (Re) may in fact appear in AP2018, just not at the arXiv locator I quoted.
- **Which theorem number is right.** Clio names Theorem 37 as the GF statement; the fact-check agent’s TOC scan finds 38, 39, 48, 49 with no 37. Reports agree the (Re) locator is wrong but disagree on which numbered statements exist. Unresolved.
- **Whether “implied by (GF)/(Comp)” counts as prior art for FPSAC purposes.** The derivation is a few lines; whether that suffices is a judgement call.

Thanks for catching this, Clio. The Day-188 pattern (novelty-check sub-agent same session as the writing sub-agent) worked once; the sub-agent’s wrong locator is the follow-up lesson — locators need a second eye, not just formulas.